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(54) **MOBILE NETWORK OPERATOR (MNO)
CONTROL OF WIFI QOS VIA
EAP/DIAMETER**

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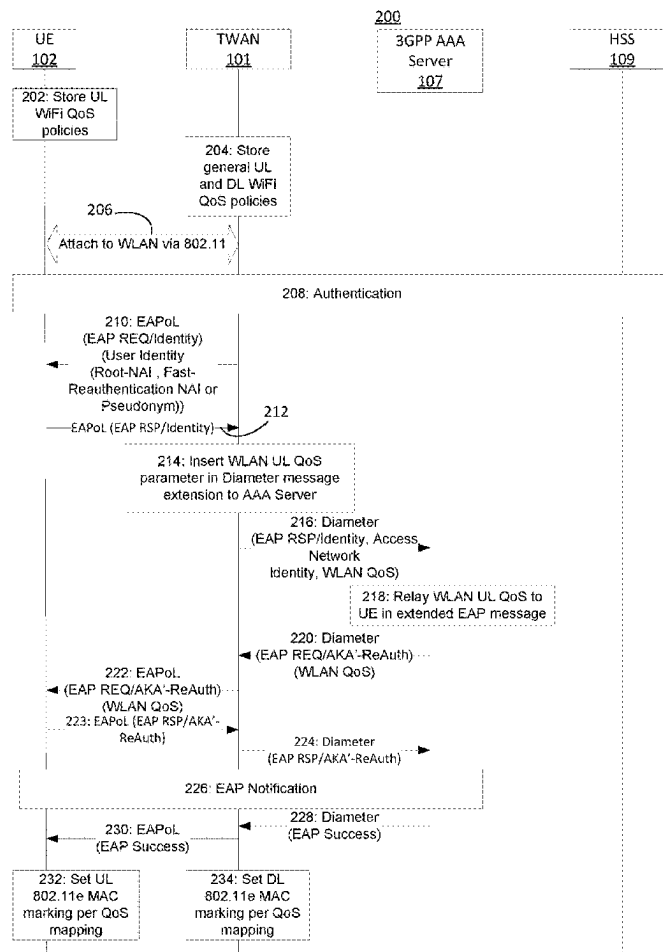
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(57) **ABSTRACT**

A mobile network operator (MNO) may control WiFi QoS. 3GPP has specified control mechanisms for various levels of quality of service (QoS) over the cellular access and core network. Embodiments described herein provide differentiation of WiFi QoS based on MNO requirements. In particular, extensible authentication protocol (EAP) and diameter messages may be extended to include a wireless local area network QoS parameter. This may be used by user equipment to set the uplink 802.11e User Priority (UP) for offloaded or evolved packet core-routed WiFi traffic.



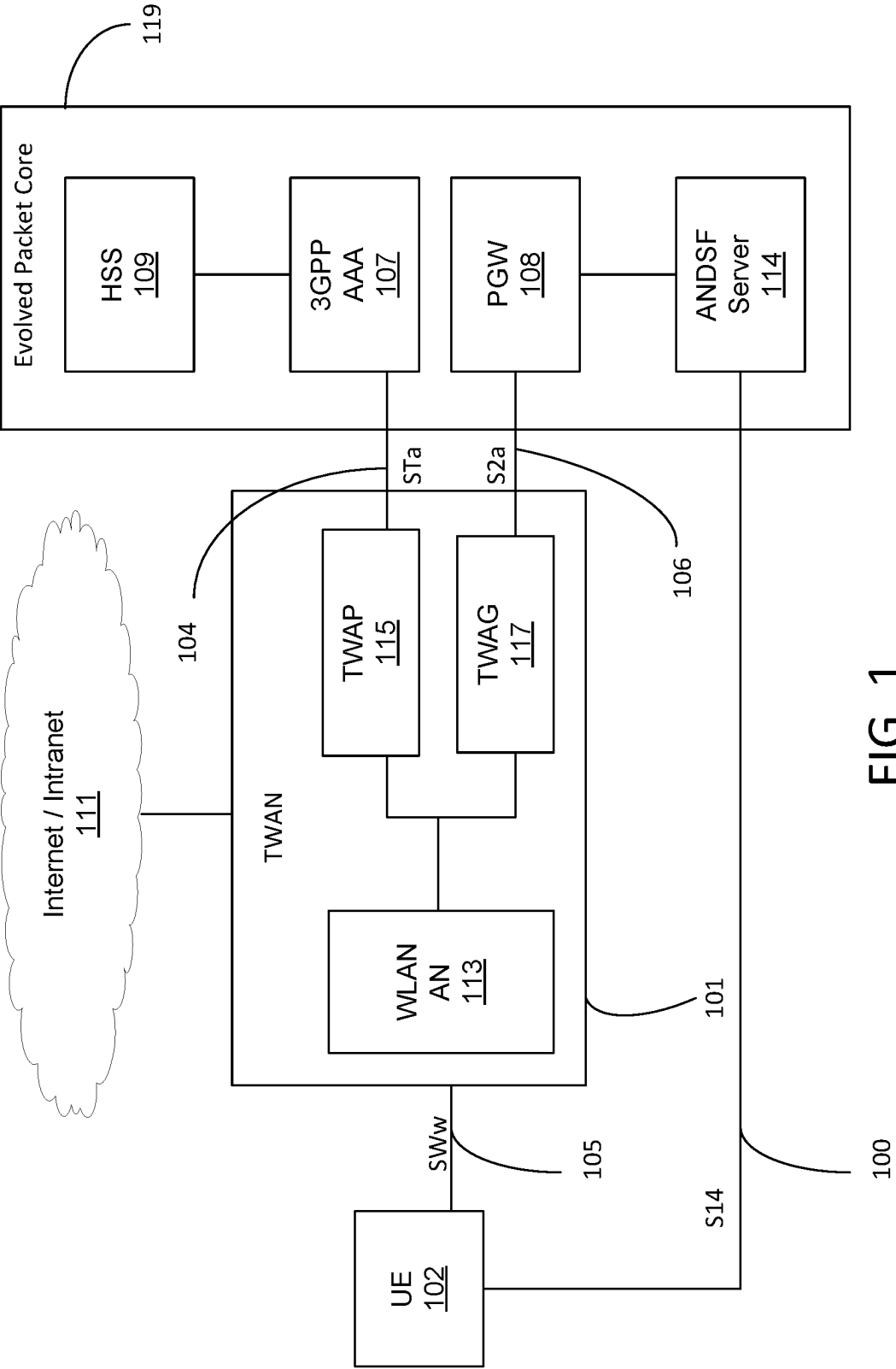


FIG. 1

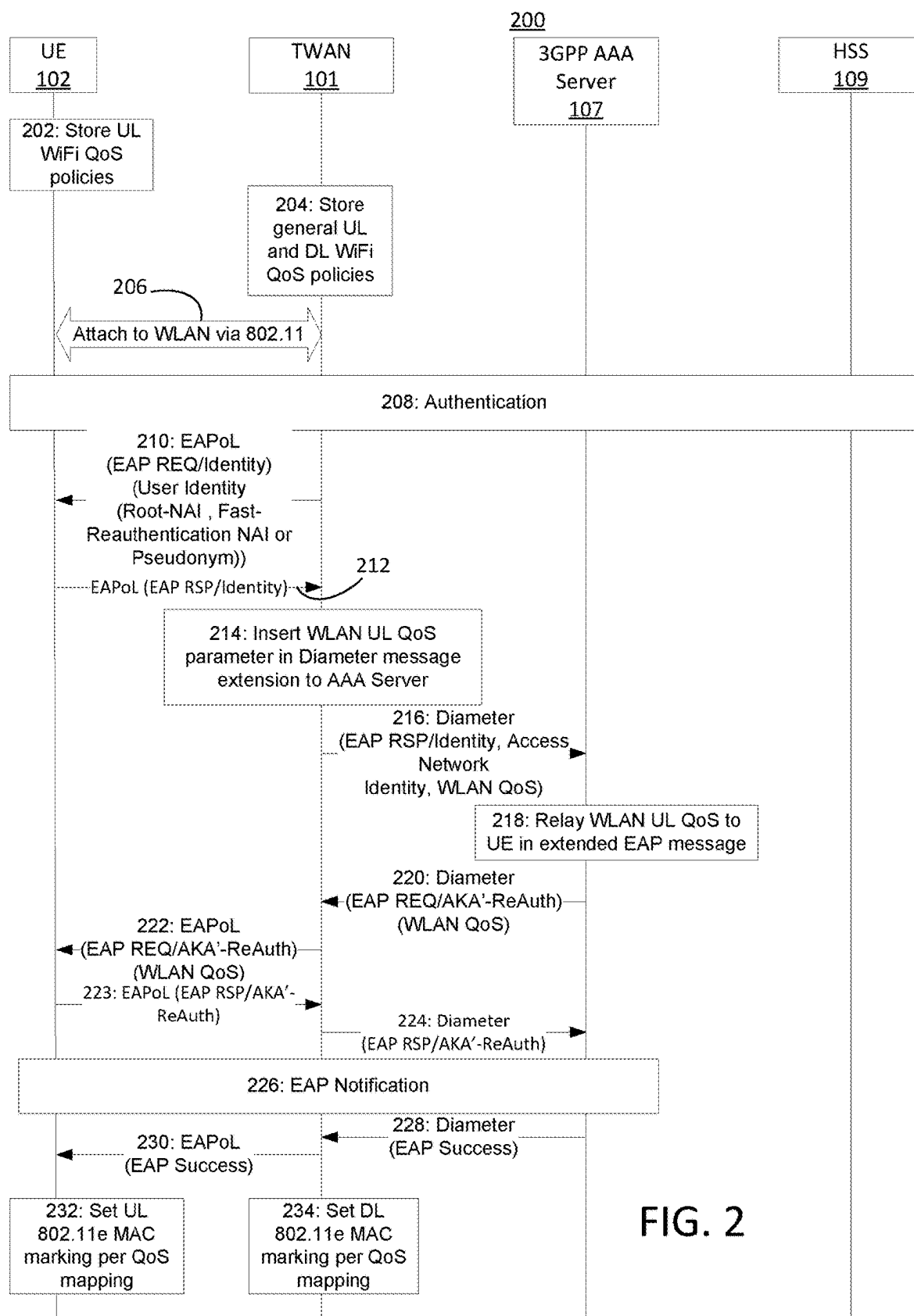


FIG. 2

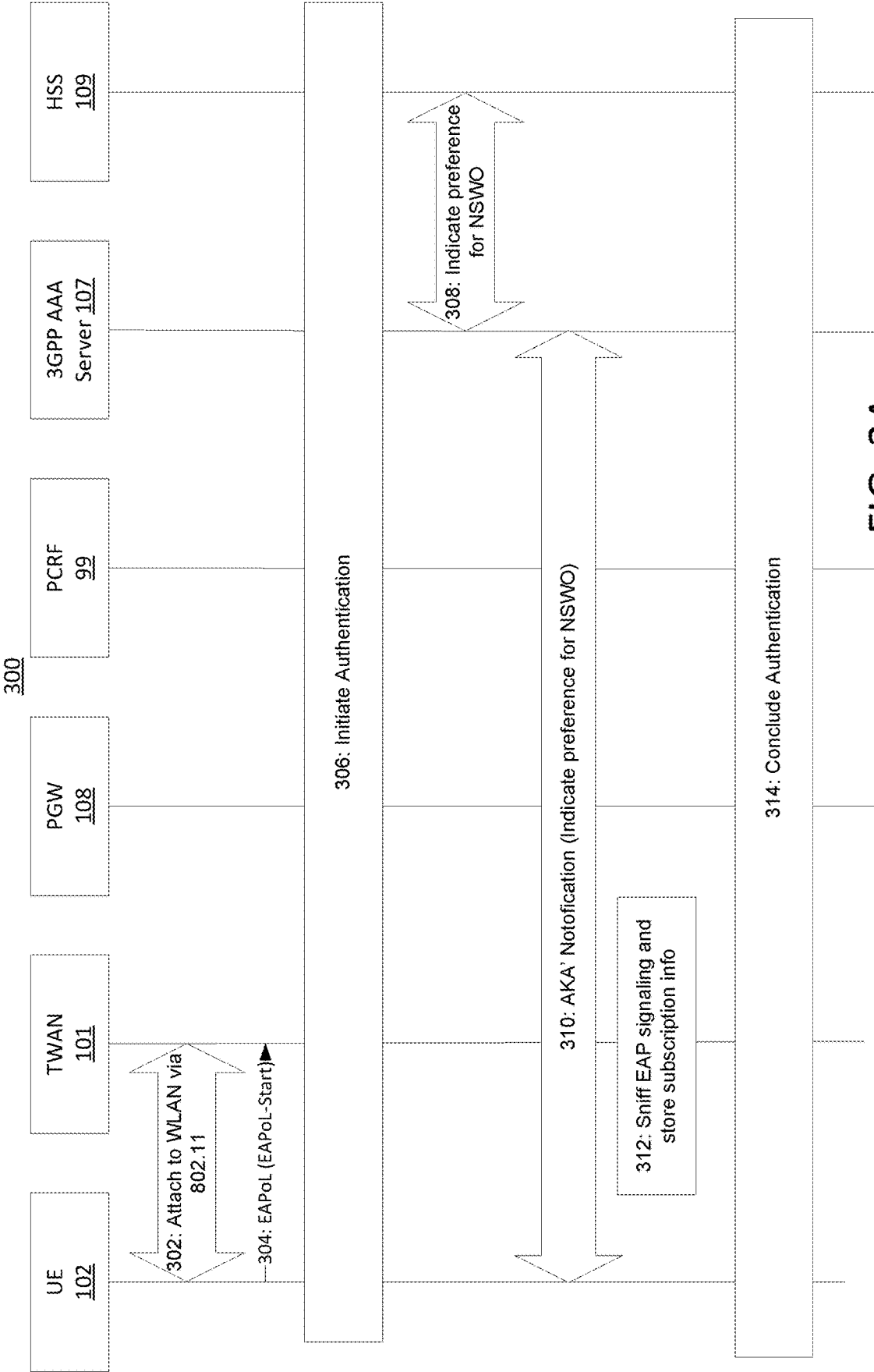
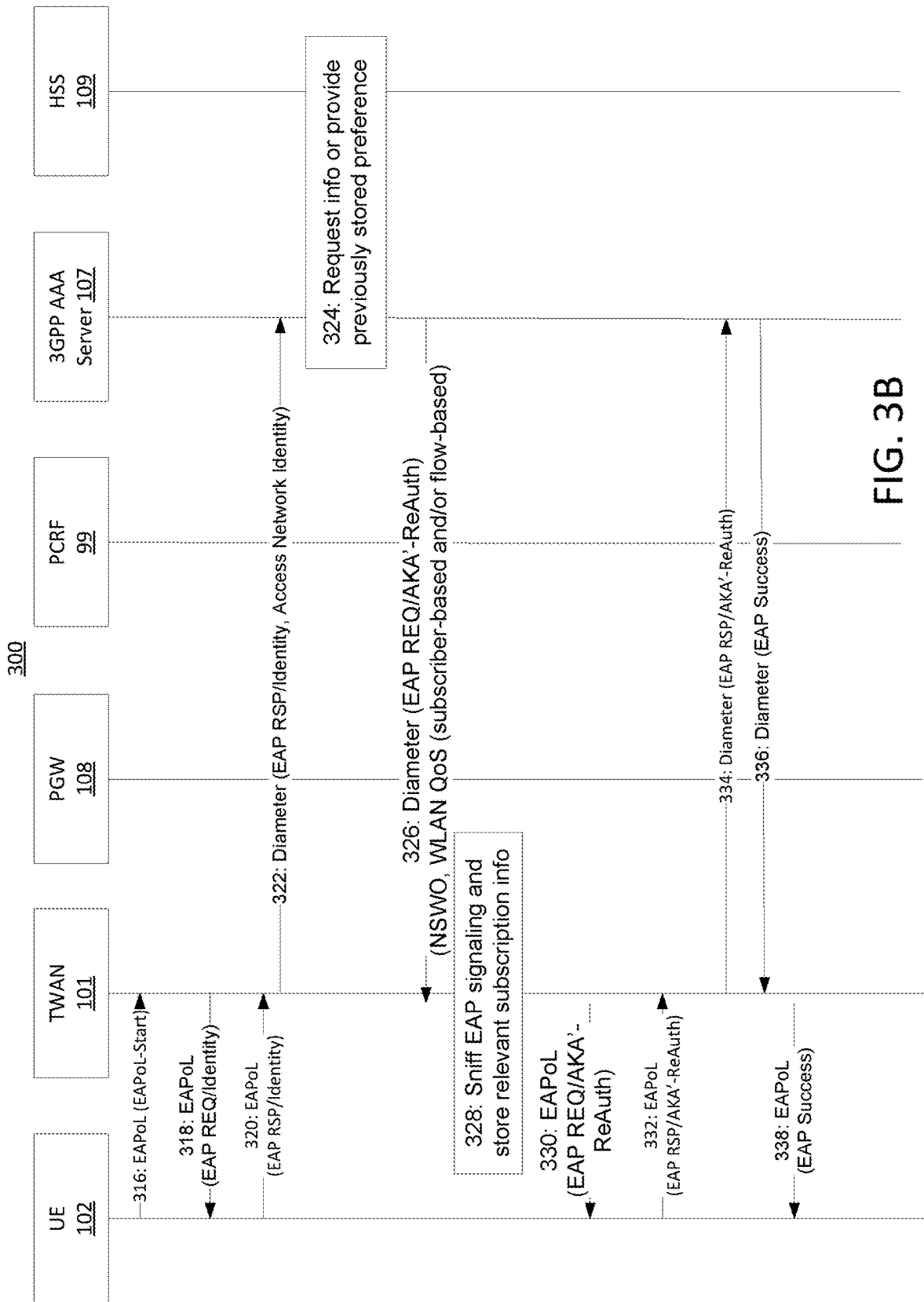
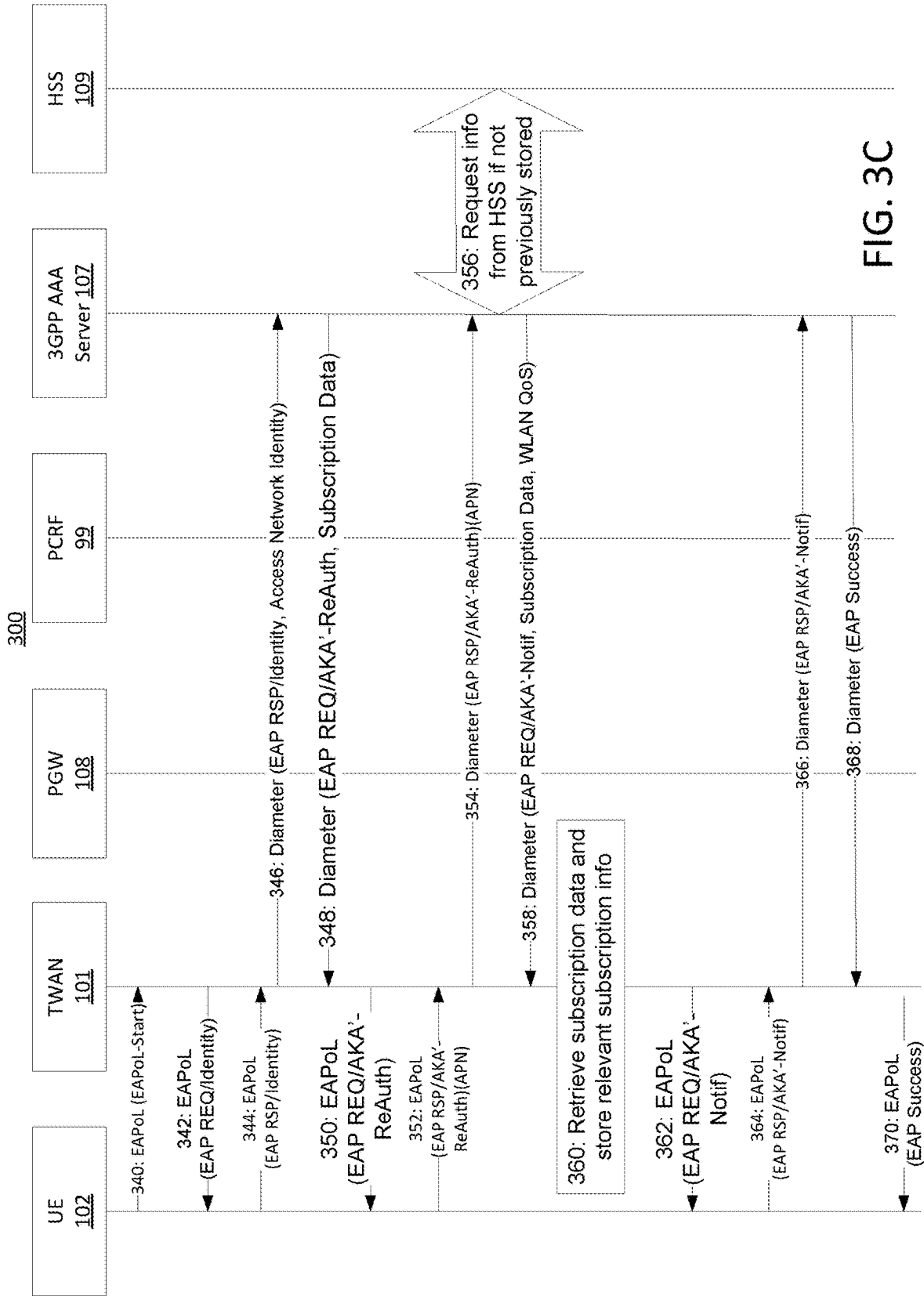


FIG. 3A





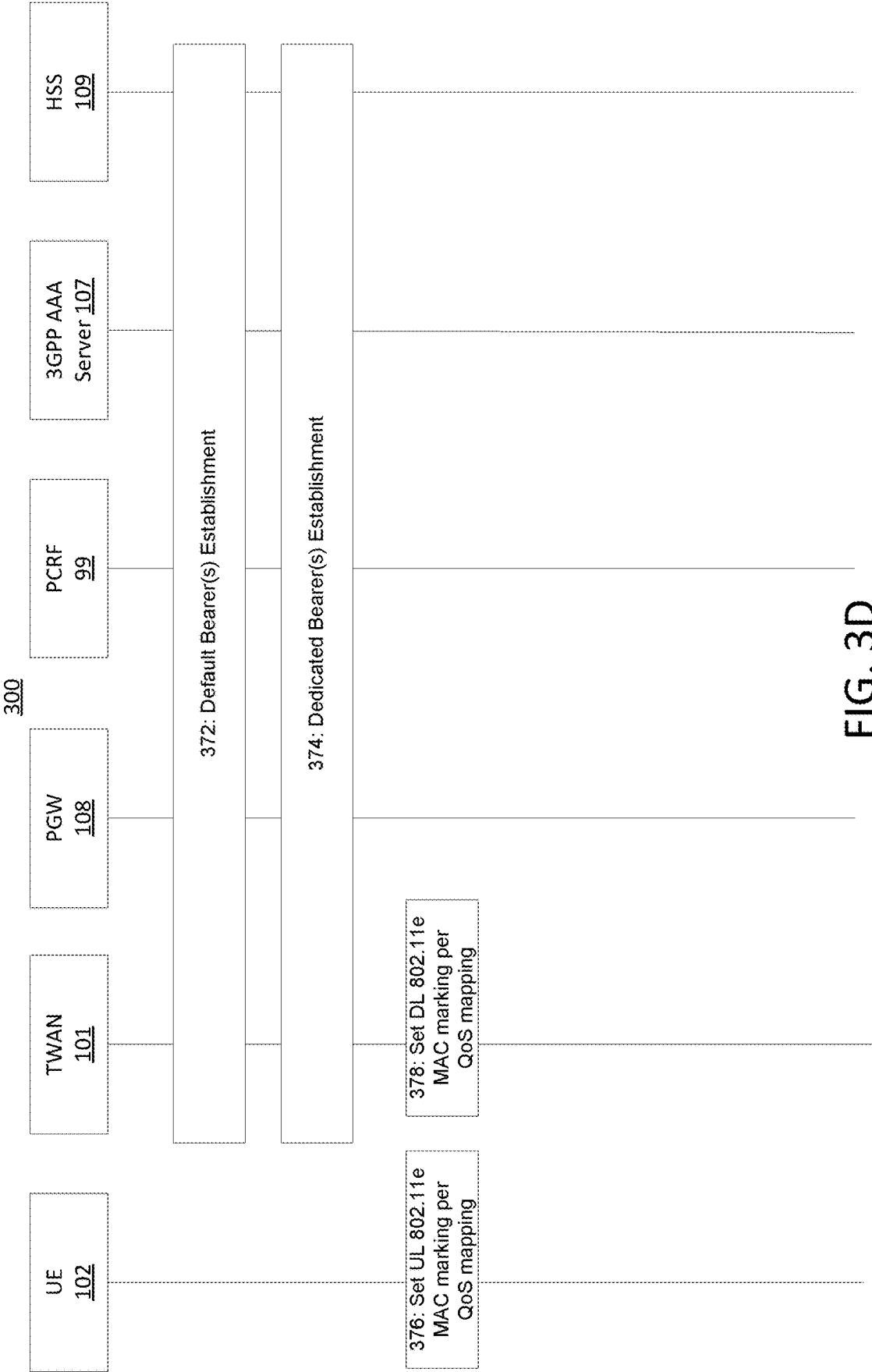


FIG. 3D

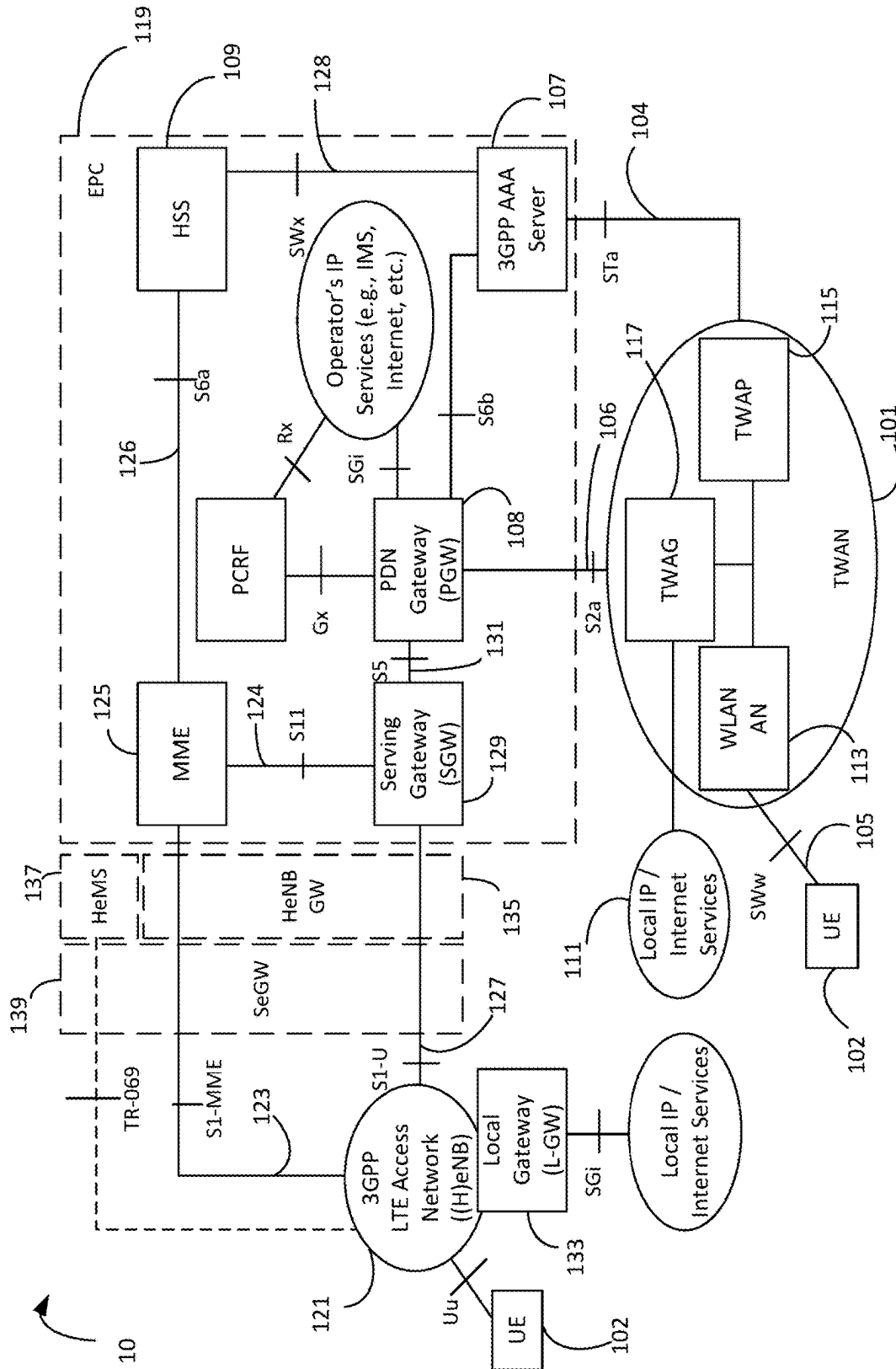


FIG. 4A

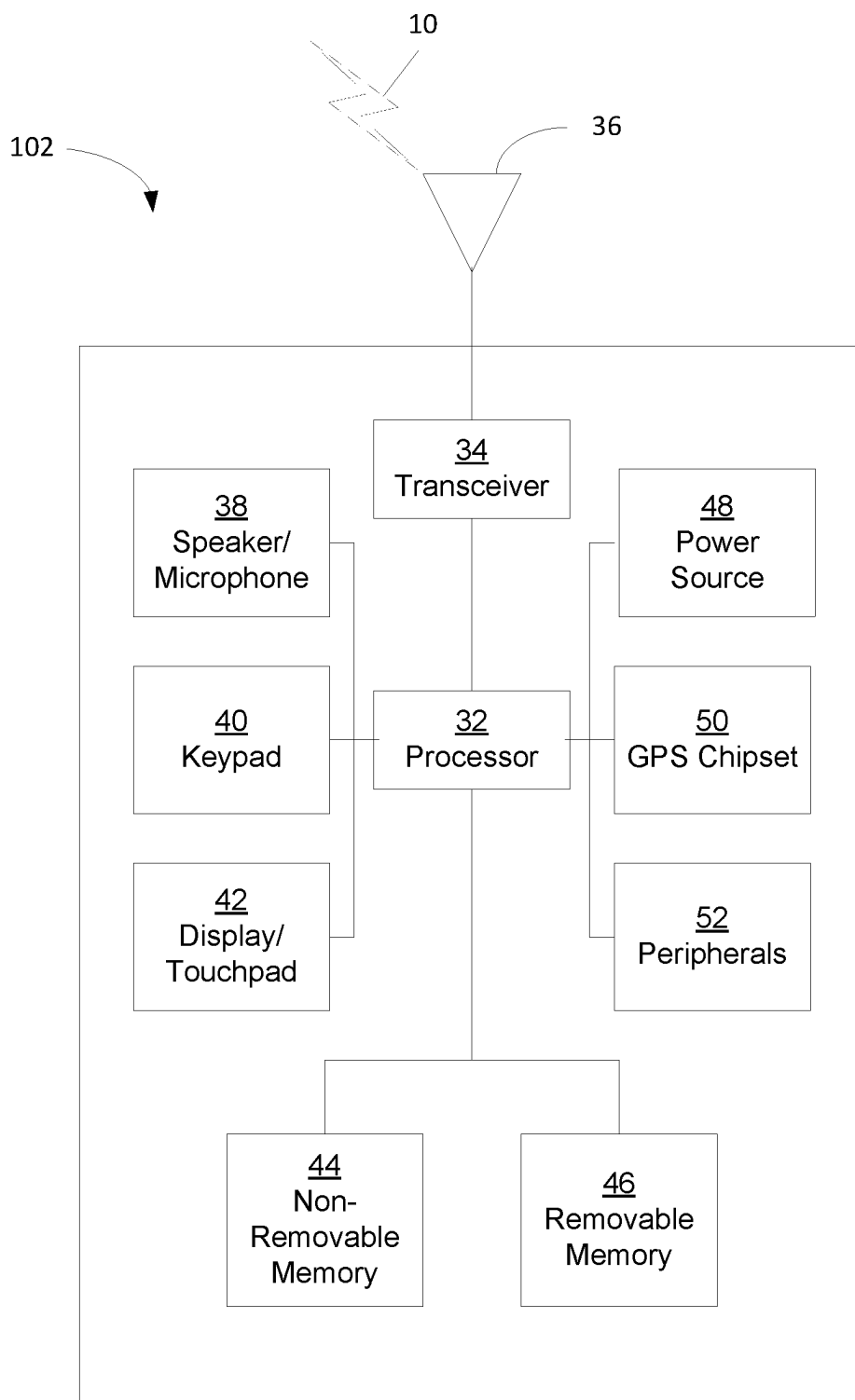


FIG. 4B

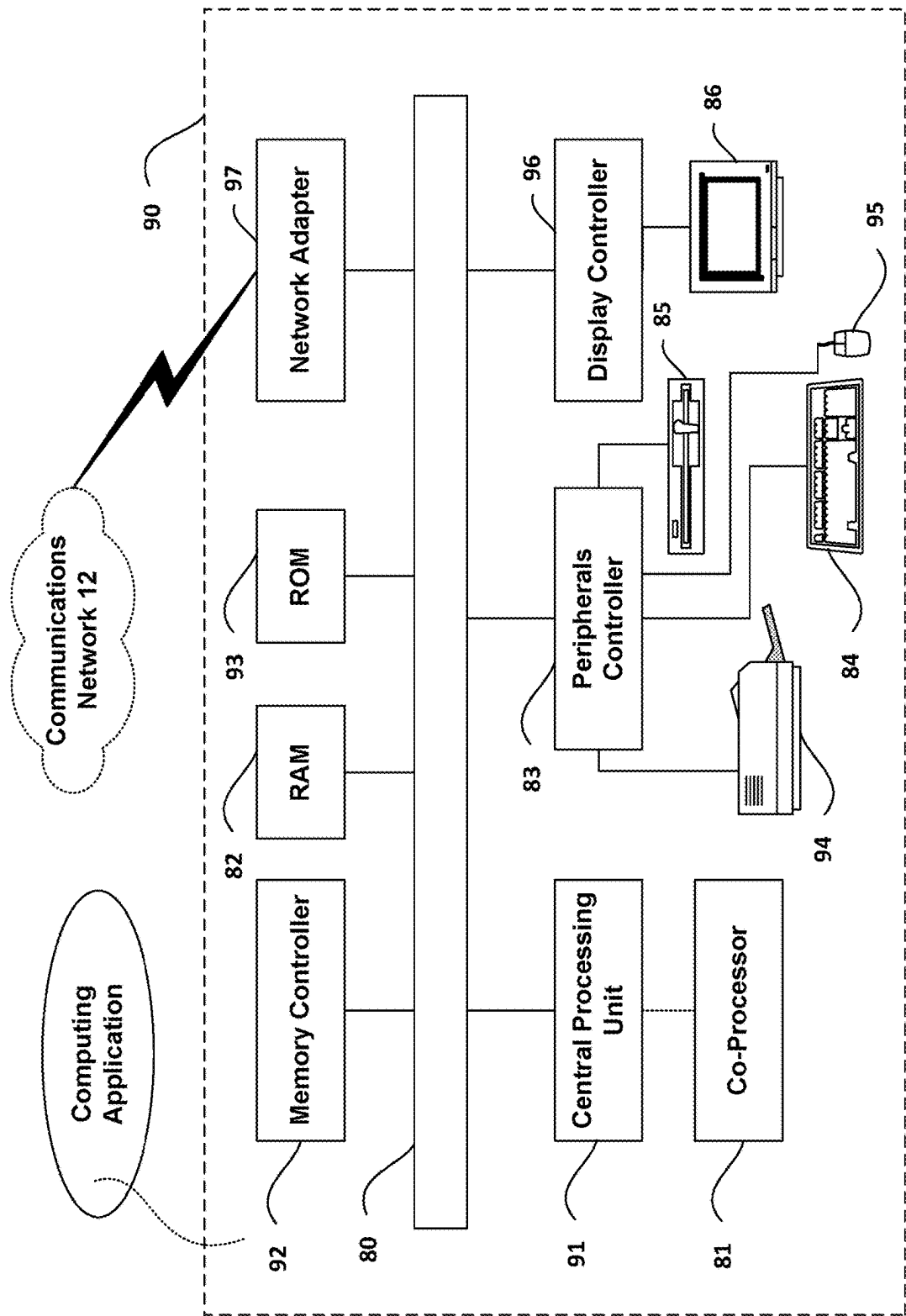


FIG. 4C

MOBILE NETWORK OPERATOR (MNO) CONTROL OF WIFI QOS VIA EAP/DIAMETER

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application Ser. No. 61/878,260, filed Sep. 16, 2013, the disclosure of which is hereby incorporated by reference as if set forth in its entirety herein.

BACKGROUND

[0002] 3GPP standards for cellular and WiFi interworking have been available since Release 6, evolving in Release 8, and expanding in subsequent releases. Institute of Electrical and Electronics Engineers (IEEE) has also ratified the 802.11u standard, Amendment for “Interworking with External Networks” which addresses wireless local area network (WLAN) enhancements in subscription service provider networks (SSPNs), such as those operated by 3GPP MNOs. However, feature deployment has been limited even though these standards enable access to operator-provided services and mobility between cellular and WiFi.

[0003] Mobile network operators (MNOs) may provide subscribers with managed network access using both cellular and WiFi technologies. Currently, MNOs typically consider WiFi only as a way to offload Internet-based traffic for their dual-mode subscribers. In current approaches, an MNO may configure certain handset applications to always use WiFi for Internet access when WiFi is available. For example, an MNO may configure applications to use WiFi when the applications are in low mobility scenarios and while they are within a WiFi hotspot. Current approaches to using WiFi may reduce congestion on the MNO’s cellular and core networks, but the reduced congestion may come with a cost.

SUMMARY

[0004] Described herein are methods, devices, and systems for MNO control of WiFi quality of service (QoS). In one embodiment in which a system comprises a user equipment (UE) and an mobile network operator (MNO), a QoS parameter may be inserted into a diameter message, whereby the UE is provided with QoS policies that are specified by the MNO. In accordance with another embodiment, a UE sets a user priority for wireless local area network uplink traffic flows according to a policy of the MNO. For example, the policy may be pre-configured or may be signaled by mechanisms described herein. Different WiFi QoS may be provided based on MNO requirements.

[0005] In an example embodiment, a server, for instance a trusted wireless access network (TWAN) server, may receive a message that indicates a wireless local area network (WLAN) quality of service (QoS) parameter. The message may be formatted in accordance with an extended diameter message. The server may insert the WLAN QoS parameter into an extended extensible authentication protocol (EAP) message, thereby providing a QoS policy specified by an MNO to the UE. Further, the TWAN server may send a diameter message to a second server. The diameter message may be indicative of an identity of the UE. The TWAN server may monitor extensible authentication protocol (EAP) messages between the UE and the second

server. Based on the monitored EAP messages and the identity of the UE, the TWAN server may identify the WLAN QoS policy associated with the UE. The TWAN server may send the identified WLAN QoS policy to the UE, for instance via an extended EAP message.

[0006] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to limitations that solve any or all disadvantages noted in any part of this disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] A more detailed understanding may be had from the following description, given by way of example in conjunction with accompanying drawings wherein:

[0008] FIG. 1 illustrates an architecture for trusted wireless local area network (WLAN) access to evolved packet core (EPC);

[0009] FIG. 2 is a flow diagram for controlling a WiFi quality of service (QoS) based on global policy in accordance with an example embodiment;

[0010] FIGS. 3A-D is a flow diagram that for controlling a WiFi QoS based on subscriber policy in accordance with example embodiments;

[0011] FIG. 4A illustrates an example architecture for trusted wireless local area network (WLAN) access and LTE EPC in which one or more disclosed embodiments may be implemented;

[0012] FIG. 4B is a system diagram of an example user equipment (UE) or other device that may be used within the communication system illustrated in FIG. 4A; and

[0013] FIG. 4C is a block diagram of an example computing system in which aspects of the communication system of FIG. 4A may be embodied.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0014] The ensuing detailed description is provided to illustrate exemplary embodiments and is not intended to limit the scope, applicability, or configuration of the invention. Various changes may be made in the function and arrangement of elements and steps without departing from the spirit and scope of the invention.

[0015] Integrated small cell and WiFi (ISW) networks are deployments of small cells in the licensed spectrum along with WiFi access points in the unlicensed spectrum. Mobile Network Operators (MNOs) are beginning to incorporate “carrier-grade” WiFi in ways that complement their cellular and core networks through cost-effective integration and interworking. This may drive development of a variety of network architectures, subscriber service options, and policy management mechanisms.

[0016] ISW network requirements may address lower cost alternatives for Internet traffic offload via WiFi, service continuity between cellular and WiFi, simplified network deployment and management (e.g., via consolidation of cellular and WiFi provisioning mechanisms and self-organizing network (SON) extensions), and enhanced policy-based multi-access traffic management (e.g., via dynamic

traffic steering and local enforcement of quality of service (QoS) across cellular and WiFi access technologies).

[0017] Disclosed herein are solutions for MNO control of WLAN QoS via Extensible Authentication Protocol (EAP) and Diameter messages. If MNOs deploy carrier WiFi, there may be a desire for access points (APs) and subscriber handsets to support at least some 802.11e or WiFi Alliance (WFA) Wireless Multimedia (WMM) QoS features so that the MNOs can offer value-added services via different levels of QoS over WiFi. For uplink data (WiFi transmission from the UE) a mechanism can be defined for providing operator-specified QoS policies to the user equipment (UE) directly from a 3GPP AAA server. The UE may also use these MNO policies to request a specified QoS level for specific down-link traffic streams from a wireless local area network (WLAN) AP.

[0018] To give further context, discussed below are relevant WiFi standards for WLANs QoS and WLANs as associated with 3GPP. 3GPP has specified control mechanisms for various levels of QoS over the cellular access and core network. As detailed herein, these capabilities are extended to include differentiation of WLAN QoS based on MNO requirements as may be similarly applied to cellular access networks.

[0019] WiFi may be used for inexpensive delivery of MNO value-added packet data services, including mobile session continuity, via unlicensed spectrum. Adjustments may be made for delivering better QoS for offloaded WiFi traffic depending on how and where the offload is done. For instance, WLANs may support QoS differentiation using the WMM standards based on IEEE 802.11e. IEEE 802.11e refers to Amendment 8: Medium Access Control (MAC) Quality of Service Enhancements. Applications using the WMM APIs can prioritize 802.11e MAC frames according to user priorities (UPs) mapped to access categories (ACs) for voice, video, best effort, or background. The four AC queues allow higher priority frames to be transmitted with a statistically lower wait time than lower priority frames.

[0020] FIG. 1 illustrates a simplified architecture for a Trusted WLAN access network (TWAN) connected with an EPC. Further details regarding the TWAN are discussed with regard to FIG. 4A thru FIG. 4C. According to section 16.1.1 of TS 23.402, when the WLAN is considered trusted by the operator, TWAN 101 interfaces with EPC 119 in multiple ways. TWAN 101 may interface with EPC 119 via STa interface 104 for authentication procedures with 3GPP AAA server 107. STa interface 104 securely transports access authentication, authorization, mobility parameters and charging-related information. In addition, TWAN 101 may interface with EPC 119 via S2a interface 106 for bearer management procedures with packet data network (PDN) gateway (PGW) 108. ANDSF server 114 may be located in EPC 119 and communicate with UE 102 via communicatively connected PGW 108. ANDSF server 114 may communicate to UE 102 using an s14 interface 100. ANDSF server 114 may initiate a push to distribute its information to UE 102 or UE 102 may query ANDSF server 114 to pull desired information.

[0021] TS 23.402 considers the detailed functional split within TWAN 101 as out of scope for 3GPP. The external behavior exposed by the SWw interface 105, S2a interface 106, and STa interface 107 are considered in scope for 3GPP. Nevertheless, functions such as WLAN AN 113, Trusted WLAN Access Gateway (TWAG) 117, and Trusted WLAN

AAA Proxy (TWAP) 115 are assumed in scope of TWAN 101. WLAN AN 113 consists of one or more WLAN Access Points (APs). An AP terminates the UE's WLAN IEEE 802.11 link via SWw interface 105. These could be stand-alone APs or APs connected to a Wireless LAN Controller (WLC), e.g., using IETF CAPWAP/DTLS protocols.

[0022] TWAG 117 acts as the default IP router for UE 102 on its access link and terminates the GTP-based S2a interface 106 with the PGW 108. It also acts as a DHCP server for UE 102. TWAG 117 maintains a UE MAC address association for forwarding packets between UE 102 and TWAG 117 via a point-to-point link through the AP (not shown) in WLAN 113 and the S2a GTP-u tunnel for UE 102 toward PGW 108. The implementation of the point-to-point link, including how and when it is setup, is out-of-scope of 3GPP (e.g., WiFi procedures are defined by the WiFi Alliance and IEEE 802.11, while WiFi network discovery and selection decisions are based on UE implementation).

[0023] TWAP 115 terminates the Diameter-based STa interface 104 with 3GPP AAA Server 107. Diameter refers to the IETF authentication, authorization, and accounting protocol. TWAP 115 relays the AAA information between WLAN AN 113 and 3GPP AAA Server 107 (or Proxy in case of roaming). TWAP 115 establishes the binding of UE 102 subscription data including international mobile subscriber identity (IMSI) with UE 102 MAC address and can inform TWAG 117 of layer 2 attach and detach events. There may be an analogy drawn to 3GPP "attach" which can be viewed as an "authentication" procedure with the core network. TWAP 115 may also provide TWAG 117 with subscription information for UE 102, such as IMSI or MAC bindings.

[0024] The 3GPP Release 11 SA2 work item for "S2a Mobility Over GTP" (SaMOG) has focused on enabling a GPRS tunneling protocol (GTP)-based S2a interface between PGW 108 and TWAN 101. The 3GPP Release 11 architectures, functional descriptions, and procedures for GTP-based S2a over Trusted WLAN access were standardized in section 16 of TS 23.402. The applicable GTP control plane protocol for tunnel management (GTPv2-C) is specified in TS 29.274 and the GTP user plane is specified in TS 29.281. A focus of SaMOG is "trusted access to the EPC," hence, the procedures begin with an "initial attachment" to EPC 101. Just as in LTE, successful completion of the initial attach procedure results in establishment of a "default" EPC 101 bearer enabling an "always-on" connection with the core network via a GTP tunnel on S2a interface 106. For SaMOG, direct offload to the Internet 111 is not relevant, because in the situation of direct offload to the Internet 111 the user plane connection to EPC 119 is bypassed and no GTP tunnels are established. Home subscriber server (HSS) 109 or 3GPP AAA server 107 may indicate via STa interface 104 whether access to EPC 119 via S2a interface 106 or the use of non-seamless WLAN offload (NSWO) or both are allowed for a subscriber.

[0025] UE 102 initiates an "initial attach" with TWAN 101 using "TWAN-specific L2 procedures" that are outside the scope of 3GPP. For WLAN, this would be via IEEE 802.11 procedures followed by the IETF EAPOL-Start message that initiates EAP procedures with the 3GPP AAA server 107 through TWAP 115. By comparison, initiation of an "initial attach" for 3GPP access is done via establishment of an RRC connection with an evolved node B (eNB) followed by

3GPP-specified non-access stratum (NAS) signaling with a mobility management entity (MME).

[0026] After standard EAP-based authentication, TWAG 115 provides TWAG 117 with the default access point name (APN) retrieved from HSS subscription data via 3GPP AAA

packet delay budget, and packet error loss rate, as shown in Table 1 which reflects information from Table 3.9, pg. 57 of “Next Generation Mobile Communications Ecosystem: Technology Management for Mobile Communications by Saad Z. Asif.

TABLE 1

Standardized QCI Characteristics					
QCI	Resource Type	Priority	Packet Delay Budget (NOTE 1)	Packet Error Loss Rate (NOTE 2)	Example Services
1 (NOTE 3)	GBR	2	100 ms	10^{-2}	Conversational Voice
2 (NOTE 3)		4	150 ms	10^{-3}	Conversational Video (Live Streaming)
3 (NOTE 3)		3	50 ms	10^{-3}	Real Time Gaming
4 (NOTE 3)		5	300 ms	10^{-6}	Non-Conversational Video (Buffered Streaming)
5 (NOTE 3)		1	100 ms	10^{-6}	IMS Signalling
6 (NOTE 4)		6	300 ms	10^{-6}	Video (Buffered Streaming) TCP-based (e.g., www, e-mail, chat, ftp, p2p file sharing, progressive video, etc.)
7 (NOTE 3)		7	100 ms	10^{-3}	Voice, Video (Live Streaming) Interactive Gaming
8 (NOTE 5)		8	300 ms	10^{-6}	Video (Buffered Streaming) TCP-based (e.g., www, e-mail, chat, ftp, p2p file sharing, progressive video, etc.)
9 (NOTE 6)		9			

[1] (NOTE 1):

A delay of 20 ms for the delay between a PCEF and a radio base station should be subtracted from a given PDB to derive the packet delay budget that applies to the radio interface. This delay is the average between the case where the PCEF is located “close” to the radio base station (roughly 10 ms) and the case where the PCEF is located “far” from the radio base station, e.g. in case of roaming with home routed traffic (the one-way packet delay between Europe and the US west coast is roughly 50 ms). The average takes into account that roaming is a less typical scenario. It is expected that subtracting this average delay of 20 ms from a given PDB will lead to desired end-to-end performance in most typical cases. Also, note that the PDB defines an upper bound. Actual packet delays-in particular for GBR traffic-should typically be lower than the PDB specified for a QCI as long as the UE has sufficient radio channel quality.

[2] (NOTE 2):

The rate of non congestion related packet losses that may occur between a radio base station and a PCEF should be regarded to be negligible. A PELR value specified for a standardized QCI therefore applies completely to the radio interface between a UE and radio base station.

[3] (NOTE 3):

This QCI is typically associated with an operator controlled service, i.e., a service where the SDF aggregate’s uplink/downlink packet filters are known at the point in time when the SDF aggregate is authorized. In case of E-UTRAN this is the point in time when a corresponding dedicated EPS bearer is established/modified.

[4] (NOTE 4):

If the network supports Multimedia Priority Services (MPS) then this QCI could be used for the prioritization of non real-time data (i.e. most typically TCP-based services/applications) of MPS subscribers.

[5] (NOTE 5):

This QCI could be used for a dedicated “premium bearer” (e.g. associated with premium content) for any subscriber/subscriber group. Also in this case, the SDF aggregate’s uplink/downlink packet filters are known at the point in time when the SDF aggregate is authorized. Alternatively, this QCI could be used for the default bearer of a UE/PDN for “premium subscribers”.

[6] (NOTE 6):

This QCI is typically used for the default bearer of a UE/PDN for non privileged subscribers. Note that AMBR can be used as a “tool” to provide subscriber differentiation between subscriber groups connected to the same PDN with the same QCI on the default bearer.

server 107. TWAG 117 then selects PGW 108 associated with the APN and sends a GTP-C “Create Session Request” to PGW 108. This request identifies the RAT type as “Non-3GPP” and includes the Default EPS Bearer QoS (as passed down from HSS 109) and a GTP Tunnel Endpoint Identifier (TEID) for TWAN 101. Note that this QoS is applicable to the GTP tunnel between the TWAG 117 and PGW 108 (S2a interface 106)—not to the actual end-to-end EPS bearer which includes the WiFi link, where the WLAN radio interface is considered out of scope for 3GPP. The default bearer QoS includes a QoS Class Identifier (QCI) for a non-guaranteed bit rate (non-GBR). The QCI value represents a resource type (GBR or non-GBR), priority level,

[0027] PGW 108 returns a “Create Session Response” to TWAG 117 including the default EPS Bearer QoS, the allocated UE 102 IP address, and a TEID for PGW 108. A GTP-U tunnel now exists between TWAG 117 and PGW 108. Packets for this EPS bearer are subsequently encapsulated with a GTPv1-U header containing the destination TEID, a UDP header identifying GTPv1-U port number 2152, and an “outer IP” header marked with DSCP values corresponding to the QCI. The DSCP mappings are established based on operator policies.

[0028] PGW 108 may also initiate creation of dedicated bearers on the GTP-based S2a interface. TWAN 101 specific

resource allocation/modification procedure may be executed in this step in order to support the dedicated bearer QoS. The details of this step are out of the scope of 3GPP.

[0029] PGW **108** may also initiate a bearer modification procedure for a GTP-based S2a bearer. This procedure is used to update the TFT for an active default or dedicated S2a bearer, or in cases when one or several of the S2a bearer QoS parameters QCI, GBR, MBR or ARP are modified (including the QCI or the ARP of the default S2a bearer), e.g. due to the HSS Initiated Subscribed QoS Modification procedure.

[0030] The IPv4 address and/or IPv6 prefix is allocated to UE **102** when a new PDN connection is established. For instance, TWAG **117** may request an IPv4 address in the GTP Create Session Request and the IPv4 address is delivered to TWAG **117** during the GTP tunnel establishment via the GTP Create Session Response from PGW **108**. When UE **102** requests the IPv4 address via DHCPv4, TWAG **117** delivers the received IPv4 address, subnet mask, default route, DNS server name, etc., to UE **102** within DHCPv4 signaling. UE **102** can use the subnet mask and the default gateway address for its packet routing decisions. Corresponding procedures are also defined for IPV6. For the case of NSWO, it is assumed TWAN **101** can support a network address translation (NAT) function and can provide the UE with a local IP address.

[0031] For Trusted WLAN access to EPC **119**, the PDN connectivity service is provided by the point-to-point connectivity between UE **102** and TWAN **101** concatenated with S2a bearer(s) between TWAN **101** and PGW **108**.

[0032] The S2a bearers include a default bearer as a minimum. When the default bearer is modified and/or when dedicated bearers are established, TFTs containing packet filters are also provided. TWAN **101** handles uplink packets based on the uplink packet filters in the TFTs received from PGW **108** for the S2a bearers of the PDN connection. Downlink packets are handled by PGW **108** based on downlink packet filters in the TFTs stored in PGW **108** for the S2a bearers of the PDN connection.

[0033] IEEE 802.11e has standardized two mechanisms for providing QoS enhancements in WLANs, namely, EDCA and HCCA. Subsequently, the WiFi Alliance has adopted some features of the 802.11e EDCA standard into their Wireless MultiMedia (WMM) certification program. The use of these standards have been limited, mostly focused on vendor-specific enterprise deployments (e.g., for voice over WLAN). It has not typically been used for interworking with 3GPP MNO QoS policies.

[0034] IEEE 802.11e includes MAC capabilities for QoS prioritization in WLANs where transmission opportunities (TXOPs) are determined based on traffic priority. Mechanisms have been standardized using a hybrid coordination function (HCF) in the AP. The HCF may be described as a “hybrid” function because it supports both 1) contention-based channel access (enhanced distributed channel access-EDCA), and 2) controlled channel access (HCF controlled channel access-HCCA). EDCA is a prioritized CSMA/CA contention-based access mechanism. EDCA maps user priorities (UP) to four “access categories” (ACs) allowing higher priority frames to be transmitted with a statistically lower wait time than lower priority frames. The backoff value for each AC is broadcast by the QoS-enabled AP in the beacon frames for use by stations in uplink transmissions. HCF Controlled Channel Access (HCCA) is a contention-

free access mechanism based on AP polling mechanisms. Although this can theoretically reduce contention on the medium, in reality there can still be uncontrollable interference from overlapping service areas.

[0035] The EDCA mechanism provides differentiated, distributed access by mapping eight different user priorities (UPs) to four access categories (ACs). The AC is derived from the UPs as shown below in Table 2, which reflects information from Table 9-1 of IEEE Std 802.11™-2012.

TABLE 2

UP-to-AC mappings				
Priority	UP (Same as 802.1D user priority)	802.1D designation	AC	Designation (inform- ative)
Lowest	1	BK	AC_BK	Background
	2	—	AC_BK	Background
	0	BE	AC_BE	Best Effort
	3	EE	AC_BE	Best Effort
	4	CL	AC_VI	Video
	5	VI	AC_VI	Video
Highest	6	VO	AC_VO	Voice
	7	NC	AC_VO	Voice

[0036] The UP values are in the range of 0-7, the same as the values defined for 802.1D user priorities (thereby simplifying the mapping). These user priorities were established for layer 2 data link frame prioritization in alignment with earlier class of service (CoS) standards including 802.1D (based on work done in 802.1p). The 802.1D designations are listed as follows: BK=Background, BE=Best Effort, EE=Excellent Effort, CL=Controlled Load, VI=Video (<100 ms delay), VO=Voice (<10 ms delay), and NC=Network Control. User priority 0 is placed into the Best Effort AC instead of the Background AC to preserve backward compatibility with non-QoS stations since the IEEE considers QoS functionality optional.

[0037] The WiFi Alliance (WFA) defined its WiFi Multi-Media (WMM) certification program called WMM-Admission Control (WMM-AC) to ensure that devices requiring QoS (e.g., for VoIP) are only admitted into the network if sufficient resources are available. For example, a WMM client can include a “Traffic Specification” (TSPEC) in a signaling request to the AP before sending traffic flows of a specific AC type, such as voice.

[0038] IEEE 802.11u has defined standards for “Interworking with External Networks” such as those managed by 3GPP MNOs. The 802.11u amendment describes methods for WLAN network discovery and selection, QoS mapping from external networks, and prioritized WLAN access for emergency services (e.g., for first responders). The WiFi Alliance has adopted some features of 802.11u network discovery and selection into their Hotspot 2.0 “Passpoint” certification program and the 802.11u QoS enhancements may be addressed in future Passpoint releases.

[0039] With respect to QoS mapping, 802.11u provides QoS mapping for subscription service provider networks (SSPNs) and other external networks that may have their own layer-3 end-to-end packet marking practice (e.g., differentiated services code point (DSCP) usage conventions).

Therefore, a way to remap the layer-3 service levels to a common over-the-air service level is necessary. The QoS map provides stations and access points with a mapping of network-layer QoS packet marking (e.g., DSCP) to 802.11e UPs.

[0040] For the downlink, at the AP, DSCP values are mapped to EDCA UPs. The non-AP station 802.11 (STA) may also use TSPEC and TCLAS elements in an add traffic stream (ADDTs) request frame to setup a traffic stream in the WLAN. In this method, the UP is specified in the traffic classification (TCLAS) element. The policy used by the AP to choose a specific method to map frames to user priorities is outside the scope of 802.11.

[0041] For the uplink, at the non-AP STA, external QoS parameters are mapped to IEEE 802.11 QoS parameters, e.g., DSCP to IEEE 802.11 UP and in turn to EDCA ACs. This mapping helps the non-AP STA to construct correct QoS requests to the AP, e.g., ADDTS Request, and to transmit frames at the correct priority. Standards do not specify how a UE sets the DSCP value for uplink packets, if at all. UE 102 may, for instance, use the value received in the corresponding downlink packet for the corresponding flow.

[0042] Table 3, which reflects information from IEEE Std 802.11™-2012 Table V-1, shows examples of differentiated services (DiffServ) per hop behavior (PHB) and DSCP mappings for 3GPP UMTS/GPRS traffic classes and 802.11e AC and UP. The mapping of the DSCP to 3GPP UMTS/GPRS traffic class is available in Global System for Mobile Association (GSMA) IR.34 v4.6, while IR.34 v9.0 adds the Evolved Packet System (EPS) QoS Class Identifier (QCI) mappings.

TABLE 3

Mapping table of DSCP to 3GPP QoS Information and EDCA ACs									
3GPP QoS Information		QoS Requirement on GPRS Roaming Exchange							UP (as in IEEE 802.11)
		DiffServ	Max	Max	MSDU	MSDU	EDCA		
Traffic Class	THP	PHB	DSCP	Delay	Jitter	Loss	Ratio	Category	802.11
Conversational	N/A	EF	101110	20 ms	5 ms	0.5%	10 ⁻⁶	AC_VO	7, 6
Streaming	N/A	AF4 ₁	100010	40 ms	5 ms	0.5%	10 ⁻⁶	AV_VI	5, 4
Interactive	1	AF3 ₁	011010	250 ms	N/A	0.1%	10 ⁻⁸	AC_BE	3
	2	AF2 ₁	010010	300 ms	N/A	0.1%	10 ⁻⁸	AC_BE	3
	3	AF1 ₁	001010	350 ms	N/A	0.1%	10 ⁻⁸	AC_BE	0
Background	N/A	BE	000000	400 ms	N/A	0.1%	10 ⁻⁸	AC_BK	2, 1

[0043] Table 4 may be constructed for EPC-based networks and reflects information from GSMA IR.34 v9.0.

TABLE 4

2G/3G/EPS QoS information and their mapping to DSCP values					
EPS					
QoS Information			IP transport		
QCI	Traffic Class	THP	Signaling indication	DiffServ PHB	DSCP
1	Conversational	N/A	N/A	EF	101110
2					
3					

TABLE 4-continued

2G/3G/EPS QoS information and their mapping to DSCP values					
EPS					
QoS Information			IP transport		
QCI	Traffic Class	THP	Signaling indication	DiffServ PHB	DSCP
4	Streaming	N/A	N/A	AF41	100010
5	Interactive	1	Yes (see note)	AF31	011010
6			No	AF32	011100
7		2	No	AF21	010010
8		3	No	AF11	001010
9	Background	N/A	N/A	BE	000000

[0044] IETF draft-kaippallimalil-netext-pmip-qos-wifi-01, “Mapping PMIP Quality of Service in WiFi Network,” outlines a recommended mapping between 3GPP QCI, DSCP, and 802.11e Access Category (AC) as shown below in Table 5.

TABLE 5

QCI	DSCP	802.11e AC	Example 3GPP service
1	EF	3 AC_VO	conversational voice
2	EF	3 AC_VO	conversational video
3	EF	3 AC_VO	real-time gaming
4	AF41	2 AC_VI	buffered streaming
5	AF31	2 AC_VI	IMS signaling

TABLE 5-continued

QCI	DSCP	802.11e AC	Example 3GPP service
6	AF31	2 AC_VI	buffered streaming
7	AF21	0 AC_BE	interactive gaming
8	AF11	0 AC_BE	web access
9	BE	1 AC_BK	e-mail

[0045] Although the WFA has adopted portions of 802.11u for network discovery and selection as part of the Hotspot 2.0 initiative and its corresponding Passpoint certification program, the QoS mapping standards have not been included to date. Hotspot 2.0 refers to an approach to public access Wi-Fi by the WFA allowing devices to automatically join a Wi-Fi subscriber service.

[0046] Given the current gaps of the conventional QoS techniques as eluded to above, there may be a need for adjustments that enable greater adoption of WLAN QoS controls, especially in light of the increased deployment of integrated small cell and WiFi networks. Defined below are extensions to EAP and Diameter for conveying uplink “WLAN QoS” parameters (e.g., QoS parameters for WiFi) to be applied by the UE. As further described below, EAP and Diameter messages may further be extended such that the messages may be interpreted by the TWAN enroute between the UE and 3GPP AAA server, thereby allowing the TWAN to set corresponding QoS for downlink traffic to the UE. Referring generally to FIGS. 2 and 3A-D, for offloaded or EPC-routed traffic, the 3GPP AAA server 107 may provide the UE 102 with uplink 802.11e user priority (UP) preferences via extended EAP signaling based on HSS 109 subscription information. The TWAN 101 may also set the downlink 802.11e UP based on information from the extended EAP and Diameter messages.

[0047] In an example embodiment, QoS levels are globally configured in the TWAN 101 (e.g., see FIG. 2). In another example embodiment, QoS levels are defined per subscriber based on information stored in the HSS 109 (e.g., see FIGS. 3A-D).

[0048] For example, a “WLAN QoS” attribute may be implemented in an EAP-AKA’ protocol. As stated in Solution 9 of 3GPP TR 23.852, which is incorporated by reference as if the disclosure of which is set forth in its entirety herein: “Between the UE and the TWAN/Authenticator, the new information is sent via EAPOL (IEEE 802.1X). Between the TWAN/Authenticator and the 3GPP AAA Server, the EAP-AKA’ payload is transported within Diameter messages. The principle is that the TWAN can read the parameters sent by the UE from the EAP messages, but cannot modify them, i.e., EAP messages are integrity protected, but not encrypted. When the TWAN needs to send a parameter to the UE, it does so indirectly by inserting the parameter in the Diameter message to the 3GPP AAA Server. Then the 3GPP AAA Server relays the parameter in the subsequent EAP message to the UE.”

[0049] Referring in particular to FIG. 2, an example system 200 includes the UE 102, the TWAN 101, the 3GPP AAA server 107, and the HSS 109. It will be appreciated that the example system 200 is simplified to facilitate description of the disclosed subject matter and is not intended to limit the scope of this disclosure. Other devices, systems, and configurations may be used to implement the embodiments disclosed herein in addition to, or instead of, a system instead of the system 200, and all such embodiments are contemplated as within the scope of the present disclosure.

[0050] Referring to FIG. 2, in accordance with the illustrated embodiment, at 202, the UE 102 may store uplink (UL) WiFi QoS policies. At 204, the TWAN 101 may store uplink and downlink (DL) WiFi QoS policies. The QoS policies, which may include parameters or attributes, may be statically configured in the TWAN 101 and be subsequently updated via an OAM server (not shown). At 206, the UE 102 may attach to the TWAN 101 via an 802.11 communication. At 208, the UE 102, TWAN 101, 3GPP AAA server 107, and HSS 109 may go through a process of authentication onto a network. Thus, the UE 102 may be authenticated at 208. At 210, the TWAN 101 may send an EAP over LAN (EAPOL) message to the UE 102. The message may include an EAP request. At 212, the UE 102 may send an EAP response to

the TWAN 101. In accordance with the illustrated embodiment, at 214, the TWAN 101, inserts one or more TWAN UL QoS parameters in a Diameter message. The one or more QoS parameters may be subsequently sent to the UE 102 in an extended EAP message. At 216, the diameter message that includes the one or more QoS parameters is sent to the 3GPP AAA server 107. The diameter message may include an AVP (“Attribute Value Pair”) parameter that conveys the TWAN QoS parameters as described herein.

[0051] Still referring to FIG. 2, in accordance with the illustrated example, at 218, the 3GPP AAA server 107 relays the TWAN UL QoS parameters to the UE 102 in an extended EAP message. For example, at 220, the 3GPP AAA server 107 may send an extended diameter message to the TWAN 101 that includes the TWAN QoS parameters. Thus, the TWAN 101, which also be referred to as a first or TWAN server 101, may receive a message that indicates a WLAN QoS parameter. The message may be formatted in accordance with a diameter message. At 222, the TWAN 101 can send an EAPOL message, in particular an EAP request message, to the UE 102 that includes the TWAN QoS parameters. Thus, the TWAN 101 may insert the WLAN QoS parameter into an extended extensible authentication protocol (EAP) message, thereby providing a QoS policy specified by the MNO to the UE 102. In accordance with the illustrated example, the QoS policy is a global policy that applies to a plurality of user equipment’s in the TWAN 101. In response to the request, the UE 102 may send an EAP response message to the TWAN 101, at 223. At 226, the UE 102, TWAN 101, and 3GPP AAA server may perform an EAP notification. At 228, in accordance with the illustrated example, a diameter message, in particular an EAP success message, is sent to the TWAN 101 from the 3GPP AAA server 107. At 230, an EAPOL message, in particular the EAP success message, is sent to the UE 102 from the TWAN 101. At 232, the UE 102 may set an UL 802.11e MAC marking per the QoS mapping. Thus, the UE 102 may set a user priority for WLAN uplink traffic flows according to the WLAN QoS policy. Thereafter, the UE 102 may provide data to the TWAN 101 based on the WLAN QoS policy. Similarly, at 234, the TWAN 101 may set a DL 802.11e MAC marking per the QoS mapping. Thus, the TWAN 101 may set a user priority for WLAN downlink traffic flows in accordance with the WLAN QoS policy. Thereafter, the TWAN 101 may provide data to the UE 102 in accordance with the WLAN QoS policy. The HSS 109 may store QoS policies specified by the MNO.

[0052] As discussed herein, it should be understood that the entities performing the steps illustrated in FIG. 2 are logical entities that may be implemented in the form of software (e.g., computer-executable instructions) stored in a memory of, and executing on a processor of, a device, server, or computer system such as those illustrated in FIG. 4B or FIG. 4C. That is, the method(s) illustrated in FIG. 2 may be implemented in the form of software (e.g., computer-executable instructions) stored in a memory of a computing device, such as the device or computer system illustrated in FIG. 4B or FIG. 4C, which computer executable instructions, when executed by a processor of the computing device, perform the steps illustrated in FIG. 2.

[0053] Referring now to FIG. 3A-D, an example system 300 includes the UE 102, the TWAN 101, the 3GPP AAA server 107, the HSS 109, the PGW 108, and an policy and charging rules function (PCRF) 99. It will be appreciated

that the example system 300 is simplified to facilitate description of the disclosed subject matter and is not intended to limit the scope of this disclosure. Other devices, systems, and configurations may be used to implement the embodiments disclosed herein in addition to, or instead of, a system instead of the system 300, and all such embodiments are contemplated as within the scope of the present disclosure. In accordance with the illustrated embodiment, the 3GPP AAA Server 107 may signal the QoS preference to the TWAN 101 based on the subscriber-specific WiFi QoS information provided by the HSS 109. As further described below with reference to FIGS. 3A-D, subscriber-specific QoS parameters may be stored in the HSS 109, conveyed to the 3GPP AAA Server 107, and sent to the UE 102 via the 3GPP AAA Server 107 using extensions to the EAP protocol as described herein. Further, the parameters, and thus the messages, may be monitored by the TWAN 101 enroute to the UE 102.

[0054] Referring in particular to FIG. 3A, at 302, in accordance with an example embodiment, the UE 102 may attach to the TWAN 101 via an 802.11 communication. At 304, the UE 304 may send an EAPOL start message to the TWAN 101. In response, at 306, an authentication may be initiated. At 308, the 3GPP AAA server 107 may indicate a preference for a non-seamless WLAN offload (NSWO). The NSWO preference may be based on a WLAN QoS policy for a subscriber and/or a flow WLAN QoS policy. At 310, the 3GPP AAA Server 107 sends an AKA' notification to the UE 102. The notification may include, and thus may indicate, the preference for NSWO, which may be per a subscriber WLAN QoS and/or a flow WLAN QoS. At 312, in accordance with the illustrated embodiment, the TWAN 101, and in particular a signaling sniffer of the TWAN 101, sniffs (interprets) EAP signaling and stores relevant subscription information at the TWAN 101. For example, TWAN 101 may monitor EAP messages between the UE 102 and the 3GPP AAA server 107, which may also be referred to as a second server. Example subscription information includes, without limitation, an NSWO preference and a WLAN QoS policy. The Signaling Sniffer may be implemented as a separate logical function while being implemented as part of the TWAN 101. At 314, the authentication, for instance the authentication of the UE 102, concludes.

[0055] Referring now to FIG. 3B, in accordance with an alternative embodiment, at 316, the UE 102 may send an EAPOL start message to the TWAN 101. In response, the TWAN 101 may send an EAPOL request message to the UE 102 (at 318). At 320, the UE 102 may send an EAPOL response message to the TWAN 101. At 322, the TWAN 101 may send a diameter message to the 3GPP AAA Server. The diameter message may include the identity of the UE 102 associated with the EAP response and an identity of the access network. Thus, the TWAN 101 may send a diameter message to the 3GPP AAA server 107, and the diameter message may be indicative of an identity of the UE 102. At 324, the 3GPP AAA server 107 requests information from the HSS 109. For example, the information may include a WLAN QoS policy that is specific to the UE 102 based on subscriber information associated with the UE 102. Alternatively, the 3GPP AAA server 107 may provide previously stored preferences, for instance during an example fast re-authentication, for NSWO. The information may include the QoS policy, which may be based on the subscriber (e.g., a user of the UE 102 and/or the UE 102 itself) or the flow,

for example. At 326, the 3GPP AAA server 107 sends a diameter message to the TWAN 101. The message may include the retrieved and/or stored information from 324. At 328, in accordance with the illustrated example, the TWAN 101 sniffs the EAP signaling and stores relevant subscription information in the TWAN 101. Thus, the TWAN 101 may monitor EAP messages between the UE 102 and the 3GPP AAA server 107. Based on the monitored EAP messages and the identity of the UE 102, the TWAN 101 may identify a WLAN QoS policy associated with the UE 102. Example subscription information may include, without limitation, the NSWO preference and the WLAN QoS policy, which may be specific to the UE 102 based on subscriber information associated with the UE. At 330, the TWAN 101 may send an EAPoL request message to the UE 102. The message may include the identified WLAN QoS policy. Thus, the TWAN 101 may send the identified WLAN QoS policy to the UE 102, for instance via an EAP message. The QoS policy may be retrieved via the HSS 109. The UE 102 may send an EAPOL response message to the TWAN 101, at 332. At 334, the TWAN 101 may send a diameter message that includes the EAP response to the 3GPP AAA server 107. At 336, the 3GPP AAA server 107 may send a diameter message that includes an EAP success message to the TWAN 101. At 338, the TWAN 101 may relay the EAP success message in an EAPOL message that is sent to the UE 102.

[0056] Referring now to FIG. 3C, in accordance with another alternative embodiment, at 340, the UE 102 may send an EAPOL start message to the TWAN 101. In response, the TWAN 101 may send an EAPoL request message to the UE 102 (at 342). At 344, the UE 102 may send an EAPOL response message to the TWAN 101. At 346, the TWAN 101 may send a diameter message to the 3GPP AAA Server 107. The diameter message may include an EAP request message that may include subscription data. At 350, the TWAN 101 may send an EAPOL request message to the UE 102. The UE 102 may send an EAPOL response message to the TWAN 101, at 352. At 354, the TWAN 101 may send a diameter message that includes the EAP response to the 3GPP AAA server 107. At 356, in accordance with the illustrated example, the 3GPP AAA server 107 requests (retrieves) information from the HSS 109 if the information was not previously stored. At 358, the 3GPP AAA server 107 sends a diameter message to the TWAN 101. The message may include the retrieved information from 356. The retrieved information may include, presented by way of example, subscription information, the APN identity, and the WLAN QoS, which may be subscriber based and/or flow based. At 360, the TWAN 101 retrieves subscription data from the received diameter message, and in particular the new diameter AVP. The TWAN 101 may store relevant WLAN QoS subscription information in the TWAN 101. At 362, the TWAN 101 may send an EAPOL request message to the UE 102. The EAPOL request message may include the WLAN QoS policy, which may be subscriber based and/or flow based. For example, in some cases, the WLAN QoS policy is specific to the UE 102. Thus, the TWAN 101 may also set a user priority for WLAN downlink traffic flows destined for the UE 102 in accordance with the WLAN QoS policy. The UE 102 may send an EAPOL response message to the TWAN 101, at 364. At 366, the TWAN 101 may send a diameter message that includes the EAP response to the 3GPP AAA server 107. At 368, the

3GPP AAA server **107** may send a diameter message that includes an EAP success message to the TWAN **101**. At **370**, the TWAN **101** may relay the EAP success message in an EAPOL message that is sent to the UE **102**. Thus, the UE **102** may be authenticated with a server that is controlled by an MNO. Based on the authentication, the UE **102** may receive a message that indicates a WLAN QoS policy. In various example embodiments, the message is formatted in accordance with an extended EAP message.

[0057] Referring now to FIG. 3D, after any of the authentications described above, a default bearer may be established between the TWAN **101** and the HSS **109**, at **372**. Alternatively, at **374**, a dedicated bearer may be established between the TWAN **101** and the HSS **109**. At **376**, the UE **101** may set the UL 802.11e MAC marking per the QoS mapping. Thus, the UE **102** may set a user priority for WLAN uplink traffic flows according to the WLAN QoS policy specified by the MNO. The UE **102** may thus provide data to the TWAN **101** based on the WLAN QoS policy. At **378**, the TWAN **101** may set the DL 802.11e MAC marking per the QoS mapping. Thus, the TWAN **101** may set a user priority for WLAN downlink traffic flows destined for the UE **102** in accordance with the WLAN QoS policy. The TWAN **101** may further provide data to the UE **102** in accordance with the WLAN QoS policy that may be specified by the MNO.

[0058] Thus, as described above, the TWAN **101** may include a signaling sniffer that reads the subscriber-specific “WLAN QOS” Diameter extensions provided to the TWAN **101** by the 3GPP AAA Server **107**, sniffs the subscriber-specific “WLAN QOS” EAP extensions provided to the UE **102** by the 3GPP AAA Server **107**, and/or provides the “WLAN QOS” and associated subscription information to the WiFi QoS Policy Manager.

[0059] Further, for downlink data over the SWw air interface, the TWAN **101** may set the 802.11e UP for traffic flows based on pre-configured values or based on an inspection of HSS subscriber info conveyed via the 3GPP AAA server **107** using extended EAP/Diameter signaling. Pre-configured policies may be used for handling the differentiation of downlink flows. For example, referring to FIG. 2, global flow-based policies match configured packet filters. In accordance with the description relative to FIGS. 3A-D, using EAP/Diameter with the described “WLAN QOS” extensions, the downlink traffic that is handled is tailored using the WLAN QOS information stored for the user in the HSS **109**.

[0060] In an example embodiment for uplink data over the SWw air interface, the UE **102** sets the 802.11e UP for traffic flows according to a pre-configured operator policy or as signaled by the new mechanisms described above. Pre-configured policies may be used for handling the differentiation of uplink flows, e.g., via flow-based policies matching specified packet filters. Using EAP with the described “WLAN QOS” extension, the uplink traffic handling may be tailored using the WLAN QOS information stored for the user in the HSS **109**.

[0061] Referring again to FIG. 3D, in some cases, dedicated bearers may be associated with a default PDN connection and may exhibit specific QoS requirements that cannot be adequately handled by the default connection. Packets belonging to the dedicated bearer may be distinguished via packet filters comprising a Traffic Flow Template (TFT). In the TWAN **101**, the TFTs may be provided

via GTP signaling from the PGW **108** and may be used to route uplink packets from the TWAN **101** to the PGW **108** via the associated dedicated bearer. The PGW **108** may use its TFTs to route downlink packets to the TWAN **101** via the associated dedicated bearer.

[0062] As discussed herein, it should be understood that the entities performing the steps illustrated in FIGS. 3A-D are logical entities that may be implemented in the form of software (e.g., computer-executable instructions) stored in a memory of, and executing on a processor of, a device, server, or computer system such as those illustrated in FIG. 4B and FIG. 4C. That is, the method(s) illustrated in FIGS. 3A-D may be implemented in the form of software (e.g., computer-executable instructions) stored in a memory of a computing device, such as the device or computer system illustrated in FIG. 4B or FIG. 4C, which computer executable instructions, when executed by a processor of the computing device, perform the steps illustrated in FIGS. 3A-D.

[0063] Disclosed below are more details with regard to 3GPP architecture that provides cellular LTE and Trusted WLAN access to an EPC. Under current practices, mobile network operators (MNOs) typically employ WiFi for off-loading “best effort” Internet traffic from their cellular and core networks. However, increased interest in operator deployment of “small cells” and “carrier WiFi” is expected to encourage MNOs to seek better inter-operability across local cellular and WiFi networks. Generally, “small cells” refer to localized geographic areas providing wireless network access via operator-licensed spectrum using 3GPP-defined cellular Radio Access Technologies (RATs). Although offloaded traffic is discussed herein, it is contemplated that devices that may primarily use WiFi communication may practice the WLAN QoS via EAP/Diameter as disclosed herein.

[0064] As operators adopt “carrier WiFi” to optimize their networks and reduce expenses, it is expected that there will be a greater deployment of “Trusted” WLAN Access Networks (TWAN) that can interface directly with an operator’s Mobile Core Network (MCN). Similarly, it is expected that there will be greater integration of MNO deployed small cell and WiFi access networks within common geographical areas such as high-traffic urban metropolitan hotspot locations. Such integration is motivated by the growing number of smartphones that support both cellular and WiFi access.

[0065] In this context, the term “trusted WLAN (TWAN) access” refers to the circumstances wherein appropriate measures have been taken to safeguard the EPC from access via the WLAN. Such measures are left to the discretion of the MNO and may, for example, include establishment of a tamper-proof fiber connection between the WLAN and EPC, or establishment of an IPSec security association between the WLAN and a Security Gateway at the EPC edge. In contrast, if the WLAN access is deemed “untrusted,” the WLAN may interface with an evolved Packet Data Gateway (ePDG) at the EPC edge, and the ePDG must establish an IPSec security association directly with each UE accessing the EPC through the WLAN.

[0066] FIG. 4A is a diagram of an example communication system **10** in which one or more disclosed embodiments may be implemented. FIG. 4A depicts a 3GPP architecture that provides cellular LTE and Trusted WLAN access to EPC **119**. As described in section 16.1.1 of 3GPP Technical Specification (TS) 23.402, the contents of which are hereby

incorporated herein by reference, when WLAN AN **113** is considered trusted by the operator, the Trusted WLAN Access Network (TWAN) **101** can be connected to Evolved Packet Core (EPC) **119** via the STa interface **104** toward the 3GPP AAA Server **107** for authentication, authorization, and accounting via the S2a interface **106** toward PDN Gateway (PGW) **108** for user plane traffic flows. An alternate path from TWAN **101** to a local IP network **111** (i.e., intranet) and/or directly to the Internet **111** is also shown.

[0067] 3GPP LTE access network **121** (i.e., evolved Node B) is connected to EPC **119** via S1-MME interface **123** which provides a communication path with Mobility Management Entity (MME) **125**. S1-U interface **127** provides a communication path with Serving Gateway (SGW) **129**, which interfaces with PDN Gateway (PGW) **108** via S5 interface **131**.

[0068] A “local gateway” function (L-GW) **133** provides small cell LTE access, e.g., for Home eNB (HeNB) deployments. Similarly, a “HeNB Gateway” (HeNB GW) **135** may be used to concentrate control plane signaling for multiple HeNBs toward MME **125** and could also be used to handle HeNB user plane traffic toward SGW **129**. A HeNB Management System (HeMS) **137** provides “plug-and-play” auto configuration of HeNBs based on TR-069 standards published by the broadband forum (BBF) and adopted by 3GPP. A security gateway (ScGW) **139** provides trusted access to EPC **119** via HeNB GW **135**.

[0069] WLAN AN **113** comprises one or more WLAN Access Points (APs). An AP (not shown) terminates UE **102** WLAN IEEE 802.11 link via SWw interface **156**. The APs may be deployed as standalone APs or as “thin” APs connected to a Wireless LAN Controller (WLC) using, for example, the IETF CAPWAP protocols.

[0070] TWAG **117** terminates the GTP-based S2a interface **106** with the PGW **108** and may act as the default IP router for UE **102** on its WLAN access link. It also may act as a DHCP server for UE **102**. TWAG **117** typically maintains a UE MAC address association for forwarding packets between UE **102** (via the WLAN AP) and the associated S2a interface **106** GTP-U tunnel (via the PGW **108**).

[0071] Trusted WLAN AAA Proxy (TWAP) **115** terminates the Diameter-based STa interface **104** with the 3GPP AAA Server **107**. TWAP **115** relays the AAA information between the WLAN AN **113** and 3GPP AAA Server **107** (or Proxy in case of roaming). TWAP **115** can inform TWAG **117** of the occurrence of layer 2 attach and detach events. TWAP **115** establishes the binding of UE subscription data (including IMSI) with UE MAC address and can provide such information to TWAG **117**.

[0072] In existing systems, UE **102** can leverage USIM features for both 3GPP and non-3GPP WLAN access. Processing for authentication and security is described in section 4.9.1 of 3GPP TS 23.402, the contents of which are hereby incorporated by reference in their entirety. As described therein, non-3GPP access authentication, such as that which takes place via WLAN AN **113**, defines the process that is used for access control and thereby permits or denies a subscriber from attaching to and using the resources of a non-3GPP IP access which is interworked with the EPC **119**. Non-3GPP access authentication signaling is executed between UE **102** and the 3GPP AAA server **107** and HSS **109**. The authentication signaling may pass through AAA proxies.

[0073] Trusted 3GPP-based access authentication is executed across STa interface **104**. The 3GPP based access authentication signaling is based on IETF protocols, e.g., Extensible Authentication Protocol (EAP). STa interface **104** and Diameter application are used for authenticating and authorizing UE **102** for EPC **119** access via trusted non-3GPP accesses. 3GPP TS 29.273, the contents of which are hereby incorporated by reference in its entirety, describes the standard TWAN procedures currently supported on STa interface **104**.

[0074] For EPC **119** access via GTP-based TWAN **101**, the IPv4 address and/or IPv6 prefix is allocated to UE **102** when a new PDN connection is established with EPC **119** over TWAN **101**. A separate IP address may also be allocated by the TWAN **101** for local network traffic and/or direct Internet offload.

[0075] For PDN connectivity through EPC **119** via TWAN **101**, TWAN **101** receives relevant PDN information via EAP/Diameter or WLCP signaling. TWAN **101** may request an IPv4 address for UE **102** from PGW **108** via the GTP Create Session Request. The IPV4 address is delivered to TWAN **101** during the GTP tunnel establishment via the GTP Create Session Response. When UE **102** requests an IPv4 address for PDN connectivity via DHCPv4, the TWAN **101** delivers the received IPv4 address to the UE **102** within DHCPv4 signaling. Corresponding procedures are also defined for IPV6.

[0076] For 3GPP LTE access, the UE **102** automatically triggers a PDN connection as part of its initial attachment to the EPC **119**. UE **102** may subsequently establish additional PDN connections as needed.

[0077] The primary purpose of the attach procedure is for UE **102** to register with the network in order to receive services for which it has subscribed to. The attach procedure confirms the user’s identity, identifies the services it is allowed to receive, establishes the security parameters (e.g., for data encryption), and notifies the network of the UE **102** initial location (e.g., in case it needs to be paged). Also, to support the “always-on” network connectivity expected by today’s users, the LTE standards specify establishment of a default PDN connection as part of the Attach procedure. The radio resources for this default connection may be released during periods of inactivity, however the rest of the connection remains intact and the end-to-end connection can be quickly re-established by reassigning the radio resources in response to UE **102** service requests.

[0078] When UE **102** attempts to attach to EPC **119** via an (H) eNB LTE network **121**, it first establishes an RRC connection with the (H) eNB LTE network **121** and encapsulates the Attach Request within the RRC signaling. (H) eNB LTE network **121** then forwards the attach request to MME **125** via S1-AP signaling on S1-MME interface **123**. MME **125** retrieves subscription information from HSS **109** via the S6a interface **126** in order to authenticate UE **102** and allow attachment to EPC **119**.

[0079] After successfully authenticating the UE **102**, MME **125** selects SGW **129** (e.g., based on proximity to the (H) eNB LTE network **121**), and also selects PGW **108** (e.g., based on the default APN retrieved from HSS **109** or a specific APN requested by UE **102**). MME **125** communicates with SGW **129** over S11 interface **124** and requests creation of the PDN connection. SGW **129** executes the signaling to establish a GTP user plane tunnel with the designated PGW **108** over the S5 interface **131**.

[0080] “GTP control” signaling takes place within the S1-AP protocol between the MME 125 and (H) eNB 121. This ultimately leads to the establishment of a GTP user plane tunnel on the S1-U interface 127 between (H) eNB 121 and SGW 129. The path for the PDN connection between the UE 102 and PGW 108 is thus completed through the (H) eNB 121 and SGW 129.

[0081] The end-to-end path for the PDN connection between the UE 102 and PGW 108 is thus completed through (H) eNB 121 and SGW 129.

[0082] In systems where communications take place via TWAN 101, UE 102 authentication and EPC 119 attachment is accomplished via EAP signaling between UE 102 and 3GPP AAA Server 107.

[0083] The PDN connectivity service is provided by the point-to-point connectivity between UE 102 and the TWAN 101, concatenated with S2a bearer(s) 106 between TWAN 101 and PGW 108. Unlike the LTE model, the WLAN radio resources are “always-on” from an EPC perspective. In other words, any power-saving optimizations are handled transparently using IEEE 802.11 procedures within the WLAN.

[0084] When UE 102 attempts to attach to EPC 119 via TWAN 101, it first establishes a Layer 2 connection with the WLAN AN 113 and encapsulates EAP messages within EAPOL signaling. WLAN AN 113 forwards the EAP messages to TWAN 115 which encapsulates the messages within Diameter signaling and forwards the messages to 3GPP AAA Server 107 via the STa interface 104. 3GPP AAA server 107 retrieves subscription information from the HSS 109 via the SWx interface 128 in order to authenticate UE 102 and allow attachment to EPC 119.

[0085] For 3GPP Release 11, 3GPP AAA Server 107 also provides TWAN 101 with information via STa interface 104 for establishing a PDN connection to the default PDN provisioned in the HSS 109. TWAN 101 then exercises GTP control plane (GTP-C) and user plane (GTP-U) protocols over S2a interface 106 directly toward PGW 108, thereby completing the PDN connection between UE 102 and PGW 108 through TWAN 101.

[0086] For 3GPP Release 12, the SaMOG phase-2 work item defines additional procedures for UE-initiated PDN connectivity, multi-PDN connectivity, and seamless inter-system handover. For the case of single-PDN capable TWAN scenarios, EAP extensions are defined to support UE-initiated PDN requests and seamless inter-system handover requests. For the case of multi-PDN capable TWAN scenarios, a WLAN Control Protocol (WLCP) is defined between the UE and TWAN to enable one or more UE PDN connection requests and seamless handover procedures. However, separate procedures are still utilized between the UE and 3GPP AAA Server for UE authentication.

[0087] FIG. 4B is a system diagram of an example user equipment, such as UE 102. Example user equipment (UE) includes, but is not limited to, a mobile station, a fixed or mobile subscriber unit, a pager, a cellular telephone, a personal digital assistant (PDA), a smartphone, a laptop, a netbook, a personal computer, a wireless sensor, consumer electronics, wearable devices, or the like. As shown in FIG. 4B, UE 102 may include a processor 32, a transceiver 34, a transmit/receive element 36, a speaker/microphone 38, a keypad 40, a display/touchpad 42, non-removable memory 44, removable memory 46, a power source 48, a global positioning system (GPS) chipset 50, and other peripherals 52. It will be appreciated that UE 102 may include any

sub-combination of the foregoing elements while remaining consistent with an embodiment. UE 102 may be a device that uses the disclosed systems, devices, and methods for WLAN QOS via EAP/Diameter.

[0088] The processor 32 may be a general purpose processor, a special purpose processor, a conventional processor, a digital signal processor (DSP), a plurality of microprocessors, one or more microprocessors in association with a DSP core, a controller, a microcontroller, Application Specific Integrated Circuits (ASICs), Field Programmable Gate Array (FPGAs) circuits, any other type of integrated circuit (IC), a state machine, and the like. The processor 32 may perform signal coding, data processing, power control, input/output processing, and/or any other functionality that enables the UE 102 to operate in a wireless environment. The processor 32 may be coupled to the transceiver 34, which may be coupled to the transmit/receive element 36. While FIG. 4B depicts the processor 32 and the transceiver 34 as separate components, it will be appreciated that the processor 32 and the transceiver 34 may be integrated together in an electronic package or chip. The processor 32 may perform application-layer programs (e.g., browsers) and/or radio access-layer (RAN) programs and/or communications. The processor 32 may perform security operations such as authentication, security key agreement, and/or cryptographic operations, such as at the access-layer and/or application layer for example.

[0089] The transmit/receive element 36 may be configured to transmit signals to, or receive signals from, WLAN AN 113 or (H) eNB 121. For example, in an embodiment, the transmit/receive element 36 may be an antenna configured to transmit and/or receive RF signals. The transmit/receive element 36 may support various networks and air interfaces, such as WLAN, WPAN, cellular, and the like. In an embodiment, the transmit/receive element 36 may be an emitter/detector configured to transmit and/or receive IR, UV, or visible light signals, for example. In yet another embodiment, the transmit/receive element 36 may be configured to transmit and receive both RF and light signals. It will be appreciated that the transmit/receive element 36 may be configured to transmit and/or receive any combination of wireless or wired signals.

[0090] In addition, although the transmit/receive element 36 is depicted in FIG. 4B as a single element, UE 102 may include any number of transmit/receive elements 36. More specifically, the UE 102 may employ MIMO technology. Thus, in an embodiment, the UE 102 may include two or more transmit/receive elements 36 (e.g., multiple antennas) for transmitting and receiving wireless signals.

[0091] The transceiver 34 may be configured to modulate the signals that are to be transmitted by the transmit/receive element 36 and to demodulate the signals that are received by the transmit/receive element 36. As noted above, UE 102 may have multi-mode capabilities. Thus, transceiver 34 may include multiple transceivers for enabling UE 102 to communicate via multiple RATs, such as UTRA and IEEE 802.11, for example.

[0092] Processor 32 may access information from, and store data in, any type of suitable memory, such as non-removable memory 44 and/or removable memory 46. Non-removable memory 44 may include random-access memory (RAM), read-only memory (ROM), a hard disk, or any other type of memory storage device. Removable memory 46 may include a subscriber identity module (SIM) card, a memory

stick, a secure digital (SD) memory card, and the like. In other embodiments, the processor 32 may access information from, and store data in, memory that is not physically located on UE 102, such as on a server or a home computer. The processor 32 may be configured to control lighting patterns, images, or colors on the display or indicators 42 in response to whether the WLAN QoS via EAP/Diameter in some of the embodiments described herein are successful or unsuccessful, or otherwise indicate the status of QoS or the processes for implementing QoS (e.g., FIGS. 2-3D with associated text).

[0093] The processor 32 may receive power from the power source 48, and may be configured to distribute and/or control the power to the other components in UE 102. The power source 48 may be any suitable device for powering UE 102. For example, the power source 48 may include one or more dry cell batteries (e.g., nickel-cadmium (NiCd), nickel-zinc (NiZn), nickel metal hydride (NiMH), lithium-ion (Li-ion), etc.), solar cells, fuel cells, and the like.

[0094] The processor 32 may also be coupled to the GPS chipset 50, which is configured to provide location information (e.g., longitude and latitude) regarding the current location of UE 102. It will be appreciated that UE 102 may acquire location information by way of any suitable location-determination method while remaining consistent with an embodiment.

[0095] The processor 32 may further be coupled to other peripherals 52, which may include one or more software and/or hardware modules that provide additional features, functionality and/or wired or wireless connectivity. For example, the peripherals 52 may include an accelerometer, an e-compass, a satellite transceiver, a sensor, a digital camera (for photographs or video), a universal serial bus (USB) port, a vibration device, a television transceiver, a hands free headset, a Bluetooth® module, a frequency modulated (FM) radio unit, a digital music player, a media player, a video game player module, an Internet browser, and the like.

[0096] FIG. 4C is a block diagram of an exemplary computing system 90 on which, for example, devices within or connected with the communication system 10 of FIG. 4A and FIG. 1 may be implemented. Computing system 90 may comprise a computer or server and may be controlled primarily by computer readable instructions, which may be in the form of software, wherever, or by whatever means such software is stored or accessed. Such computer readable instructions may be executed within central processing unit (CPU) 91 to cause computing system 90 to do work. In many known workstations, servers, and personal computers, central processing unit 91 is implemented by a single-chip CPU called a microprocessor. In other machines, the central processing unit 91 may comprise multiple processors. Coprocessor 81 is an optional processor, distinct from main CPU 91 that performs additional functions or assists CPU 91. CPU 91 and/or coprocessor 81 may receive, generate, and process data related to the disclosed systems and methods for WLAN QoS via EAP/Diameter, such as receiving appropriate diameter messages or EAP response or request messages.

[0097] In operation, CPU 91 fetches, decodes, and executes instructions, and transfers information to and from other resources via the computer's main data-transfer path, system bus 80. Such a system bus connects the components in computing system 90 and defines the medium for data

exchange. System bus 80 typically includes data lines for sending data, address lines for sending addresses, and control lines for sending interrupts and for operating the system bus. An example of such a system bus 80 is the PCI (Peripheral Component Interconnect) bus.

[0098] Memory devices coupled to system bus 80 include random access memory (RAM) 82 and read only memory (ROM) 93. Such memories include circuitry that allows information to be stored and retrieved. ROMs 93 generally contain stored data that cannot easily be modified. Data stored in RAM 82 can be read or changed by CPU 91 or other hardware devices. Access to RAM 82 and/or ROM 93 may be controlled by memory controller 92. Memory controller 92 may provide an address translation function that translates virtual addresses into physical addresses as instructions are executed. Memory controller 92 may also provide a memory protection function that isolates processes within the system and isolates system processes from user processes. Thus, a program running in a first mode can access only memory mapped by its own process virtual address space; it cannot access memory within another process's virtual address space unless memory sharing between the processes has been set up.

[0099] In addition, computing system 90 may contain peripherals controller 83 responsible for communicating instructions from CPU 91 to peripherals, such as printer 94, keyboard 84, mouse 95, and disk drive 85.

[0100] Display 86, which is controlled by display controller 96, is used to display visual output generated by computing system 90. Such visual output may include text, graphics, animated graphics, and video. Display 86 may be implemented with a CRT-based video display, an LCD-based flat-panel display, gas plasma-based flat-panel display, or a touch-panel. Display controller 96 includes electronic components required to generate a video signal that is sent to display 86.

[0101] Further, computing system 90 may contain network adaptor 97 that may be used to connect computing system 90 to an external communications network 10.

[0102] It is understood that any or all of the systems, methods and processes described herein may be embodied in the form of computer executable instructions (i.e., program code) stored on a computer-readable storage medium which instructions, when executed by a machine, such as a computer, server, UE, or the like, perform and/or implement the systems, methods and processes described herein. Specifically, any of the steps, operations or functions described above may be implemented in the form of such computer executable instructions. Computer readable storage media include volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information, but such computer readable storage media do not include signals. Computer readable storage media include, but are not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other physical medium which can be used to store the desired information and which can be accessed by a computer.

[0103] In describing preferred embodiments of the subject matter of the present disclosure, as illustrated in the Figures, specific terminology is employed for the sake of clarity. The claimed subject matter, however, is not intended to be

limited to the specific terminology so selected, and it is to be understood that each specific element includes all technical equivalents that operate in a similar manner to accomplish a similar purpose.

[0104] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

1.-18. (canceled)

19. A wireless transmit/receive unit (WTRU) comprising: a processor, memory, and a transceiver which are configured to:

receive an extensible authentication protocol (EAP) identity request message from a first network entity,
send an EAP identity response message, including information indicating an identity of the WTRU, to the first network entity,
perform authentication of the WTRU with a second (AAA) network entity,
receive, upon successful authentication of the WTRU, information indicating one or more wireless local area network (WLAN) quality of service (QoS) parameters associated with a flow, and
mark uplink traffic of the flow with an identifier associated with the one or more WLAN QoS parameters.

20. The WTRU of claim **19**, wherein the processor, the memory, and the transceiver are configured set a user priority for the uplink traffic of the flow according to the one or more QoS parameters.

21. The WTRU of claim **19**, wherein the processor, the memory, and the transceiver are configured to receive downlink traffic of the flow in accordance with the one or more WLAN QoS parameters.

22. The WTRU of claim **19**, wherein the uplink traffic flow is sent to the first network entity.

23. The WTRU of claim **19**, wherein the downlink traffic flow is received from the first network entity.

24. The WTRU of claim **1**, wherein the first network entity is an access point of a trusted wireless local access network (TWLAN).

25. The WTRU of claim **1**, wherein the second network entity provides a 3GPP AAA function.

26. The WTRU of claim **1**, wherein the identity request message, the EAP identity response message, and the authentication are via 802.1x communication.

27. The WTRU of claim **1**, wherein the processor, the memory, and the transceiver are configured to perform 802.1x medium access control (MAC) marking of the uplink traffic of the flow using the one or more WLAN QoS parameters.

28. The WTRU of claim **1**, wherein the identifier is a differentiated services code point (DSCP) value.

29. A method implemented by a wireless transmit/receive unit (WTRU), the method comprising:

receiving an extensible authentication protocol (EAP) identity request message from a first network entity;
sending an EAP identity response message, including information indicating an identity of the WTRU, to the first network entity;
performing authentication of the WTRU with a second (AAA) network entity,
receiving, upon successful authentication of the WTRU, information indicating one or more wireless local area network (WLAN) quality of service (QoS) parameters associated with a flow; and
marking uplink traffic of the flow with an identifier associated with the one or more WLAN QoS parameters.

30. The method of claim **29**, further comprising: setting a user priority for the uplink traffic of the flow according to the one or more QoS parameters.

31. The method of claim **29**, further comprising: receiving downlink traffic of the flow in accordance with the one or more WLAN QoS parameters.

32. The method of claim **29**, wherein the uplink traffic flow is sent to the first network entity.

33. The method of claim **29**, wherein the downlink traffic flow is received from the first network entity.

34. The method of claim **29**, wherein the first network entity is an access point of a trusted wireless local access network (TWLAN).

35. The method of claim **29**, wherein the second network entity provides a 3GPP AAA function.

36. The method of claim **29**, wherein the identity request message, the EAP identity response message, and the authentication are via 802.1x communication.

37. The method of claim **29**, further comprising: performing 802.1x medium access control (MAC) marking of the uplink traffic of the flow using the one or more WLAN QoS parameters.

38. The method of claim **29**, wherein the identifier is a differentiated services code point (DSCP) value.

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